

Cardiac Field Artifact Removal in Ear-EEG Using Repetitive Disturbance Rejection

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Abstract—Continuous wearable EEG is increasingly paired with consumer wearables that provide a heartbeat stream (e.g., smartwatch ECG/PPG). Given the available ECG/heartbeat reference, we propose a repetitive-control framework that models cardiac artifact as a heartbeat-aligned periodic disturbance. Each EEG sample is assigned a cardiac phase index wrapped around ECG R-peaks; phase-synchronous averaging then preserves the heartbeat-correlated component while attenuating non-phase-locked activity. Subtracting this phase-indexed average yields effective artifact removal in the non-causal offline setting. For online real-time operation, an iterative-learning estimator is used to track the disturbance estimate in either a same-beat predictive mode (strict causal) or a one-beat-delayed mode (buffered causal). The non-causal and buffered modes are evaluated on behind-the-ear EEG from a public wearable epilepsy dataset and on semi-synthetic data in which reference predictability is swept from linear to strongly nonlinear.

I. INTRODUCTION

Wearable EEG is moving toward continuous, day-scale monitoring, and many users already wear a second device that provides heartbeat timing from ECG or PPG. This creates a practical path for low-channel ear-EEG denoising: use heartbeat timing already available from the wearable ecosystem rather than requiring additional EEG channels. In ear-centered montages, cardiac field artifact (CFA) is often prominent because electrodes are close to neck and vascular pathways. As one of the primary contaminations in wearable ear-EEG, CFA is the volume-conducted electrical field of the heart [1], distinct from the scanner-related ballistocardiogram artifact emphasized in EEG-fMRI work [2].

Classical ECG-referenced adaptive cancellation is well established [4]–[6], and it is a natural baseline. In our target regime it faces three practical limits: earables are channel-starved for blind source separation [7], learned denoisers are benchmarked mainly on high-density montages [8], and—most restrictive for a wearable deployment—reference-driven cancellation must be driven by the ECG *waveform*, sample-aligned to the EEG. Our real-data observations sharpen this last limit rather than dismissing it: after beat-synchronous pooling, the ear-EEG artifact retains appreciable coherence with the ECG over the lower part of the band, so an

aligned waveform reference does carry usable information. The obstacle is obtaining one. A companion wearable exposes a heartbeat event stream rather than a synchronized ECG trace, wrist PPG carries no ECG waveform at all, and cross-device latency and jitter are unknown and time-varying. Beat timing, by contrast, is a single scalar per cardiac cycle that crosses the device boundary intact, and the artifact remains phase-locked to it. This motivates a timing-only control view: model cardiac contamination as a repetitive disturbance and reject it by heartbeat phase indexing. Repetitive control and iterative learning control (ILC) are mature tools for periodic disturbance rejection in mechatronics [9], and they map naturally to the heartbeat cycle when reliable beat timing is available. In this work, we focus on phase-aligned template learning/cancellation driven by heartbeat timing in three deployment modes: non-causal offline, buffered causal (one-beat latency), and strict real-time causal (zero added latency). The resulting contribution is a compact, wearable-oriented CFA removal scheme aligned with realistic device constraints: low channel count, low compute budget, and timing references that may come from a separate wearable.

The rest of this paper is organized as follows. Section II states the signal model and timing-only CFA-removal problem. Section III develops repetitive template cancellation in non-causal and causal forms. Section IV reports experiments on real ear-EEG and semi-synthetic data. Section V summarizes the main findings and practical implications.

II. PROBLEM FORMULATION

Following repetitive-control modeling conventions [9] [10], we model the measured ear-EEG signal as

$$y(t) = s(t) + d_r(t) + n(t), \quad (1)$$

where $s(t)$ is the neural signal of interest, $d_r(t)$ is the heartbeat-synchronous repetitive cardiac disturbance, and $n(t)$ aggregates asynchronous interference and sensor noise. Explicit generative models of the ECG waveform itself [11] are deliberately not required here. The subscript r denotes the repetitive nature of the disturbance. Its period is heartbeat dependent and is set by ECG R-peak timing, with beat interval

$$T_k = t_{k+1} - t_k, \quad (2)$$

where t_k is the k -th detected R-peak instant.

Assumption 1. *In ear-EEG, CFA is dominated by the repetitive component $d_r(t)$. The R-peak phase-asynchronous disturbances and noise $n(t)$ are assumed zero-mean under the*

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R-peak-synchronous averaging, and DC offset and slow drift are removed by preprocessing.

Assumption 2. We assume the neural signal can be decomposed as $s(t) = s_r(t) + s_n(t)$, where $s_r(t)$ is the heartbeat-repetitive neural component and $s_n(t)$ is the non-repetitive neural component of interest. The targeted applications in this paper do not seek to recover $s_r(t)$, so it is dropped from the problem formulation.

Using heartbeat phase, we represent the repetitive disturbance as a phase-indexed waveform

$$d_r(t) = d_r(\phi(t)), \quad \phi(t) = \frac{t - t_k}{T_k}, \quad t \in [t_k, t_{k+1}), \quad (3)$$

with $\phi(t) \in [0, 1)$ the *R*-peak-aligned cardiac phase.

Assumption 3. The beat period T_k is assumed to be accurately locked, and slowly varying due to respiratory sinus arrhythmia (RSA), with additional small modulation from low-level muscle/motion effects in resting-state recordings.

Problem 1. Find the repetitive cardiac artifact template $\hat{d}_r(\phi(t))$ such that subtracting it from the measured signal recovers the non-repetitive neural signal of interest, after dropping the out-of-scope repetitive neural component $s_r(t)$:

$$\hat{s}_n(t) = y(t) - \hat{d}_r(\phi(t)) \approx s_n(t). \quad (4)$$

Under Assumptions 1–3, $\hat{d}_r(\phi(t))$ can be estimated from the *R*-peak-locked structure of $y(t)$ alone.

Note 1. Since $d_r(\phi(t))$ is periodic in the cardiac phase $\phi(t)$ indexed solely by *R*-peak instants $\{t_k\}$, Problem 1 depends only on *R*-peak timing and never on the ECG waveform shape. Consequently, any wearable heartbeat source—smartwatch ECG, wrist PPG, or any device delivering beat timing via a synchronization mechanism—is sufficient to instantiate the solution.

Note 2. In reality, the out-of-scope $s_r(t)$ may contain a phase-synchronous component, e.g. the heartbeat-evoked potential (HEP)—a genuine cortical response time-locked to the cardiac cycle [12]. This component may be absorbed into $\hat{d}_r$; preserving it is a meaningful extension worthy of further investigation.

To instantiate Problem 1 on the discrete sample grid, let $\hat{t}_k$ be the estimated k -th *R*-peak sample instant and f_s the sampling rate. Retaining the beat duration T_k is essential: the k -th beat is first mapped from elapsed time since *R* into normalized phase:

$$d_r^{(k)}(\hat{t}_k + \phi T_k) \triangleq d_r^{(k)}(\phi), \quad (5)$$

with $\phi \in [0, 1)$ and the sampled phase map

$$\phi_{k,i} \triangleq \frac{i - \hat{t}_k}{f_s T_k}, \quad (6a)$$

$$i \in \{\hat{t}_k, \dots, \hat{t}_{k+1} - 1\}. \quad (6b)$$

Note 3. For offline non-causal processing, assuming, *R*-peaks are accurately detected and bad beats are rejected, we average

beats on a common normalized phase grid across N_r beats as:

$$\hat{d}_r(\phi) = \frac{1}{N_r} \sum_{k=1}^{N_r} y_k(\phi), \quad y_k(\phi) \equiv y(\hat{t}_k + \phi T_k). \quad (7)$$

Substituting the signal $y_k(\phi)$ by its definition from the discrete time version of Eq. (1), we obtain $\hat{d}_r(\phi) \equiv d_r(\phi)$, as $s_{n,k}(\phi)$ and $n_k(\phi)$ are averaged to zero over time. For more detailed analysis, we can refer to the established template-based BCG/CFA artifact-removal idea of [3], and we will focus on real-time solution of Problem 1 in the next sections.

III. REPETITIVE DISTURBANCE REJECTION

A. Buffered Template Averaging

For streaming template learning, the batch average of Eq. (7) is replaced by its standard exponentially-weighted ILC recursion, updated once per learnable beat at each phase ϕ :

$$\hat{d}_r^{(k)}(\phi) = \hat{d}_r^{(k-1)}(\phi) + a [y_k(\phi) - \hat{d}_r^{(k-1)}(\phi)], \quad (8)$$

where $a \in (0, 1]$ is a tunable learning gain. The supervisor freezes the update on premature/missed beats.

Note 4. In streaming operation, the averaging in Eq. (8) has to run with one beat buffered, as in strict real-time the k *R*-peak has not arrived yet. Once the averaging converges, this ILC recursion converges to the exponentially averaged normalized-phase template of Ref. [3].

B. Real-Time Cancellation

For applications where one beat of delay is acceptable, buffered cancellation is a strong candidate: the template is updated from the current learnable beat, while the preceding beat is cancelled using the completed template and its detected *R*-peak. This avoids prediction error at the cancellation anchor while adding only one heartbeat of latency:

$$\hat{s}_{k-1}(\phi) \approx y_{k-1}(\phi) - \hat{d}_r^{(k-1)}(\phi). \quad (9)$$

The buffered mode is therefore preferred whenever the application can tolerate one-beat delay.

C. Strict real-time cancellation.

When zero added latency is required, the cancellation anchor must be based on effective *R*-peak prediction rather than the detected current-beat *R*-peak. At sample i , after the last detected peak $\hat{t}_k$, the normalized predicted phase is

$$\hat{s}_k[i] = y[i] - \hat{d}_r(\hat{\phi}_{\text{pred}}(i)), \quad \hat{\phi}_{\text{pred}}(i) = \frac{i - \hat{t}_k}{f_s \hat{T}_k}. \quad (10)$$

The predictor estimates the next RR interval from a scalar random-walk model. In sample units, the observed interval z_k is the difference between consecutive detected *R*-peak sample instants, $z_k = \hat{t}_k - \hat{t}_{k-1}$.

$$T_k = T_{k-1} + w_k, \quad w_k \sim (0, q), \quad (11a)$$

$$z_k = T_k + v_k, \quad v_k \sim (0, r). \quad (11b)$$

Here q is the beat-to-beat RR-increment variance, representing rate wander such as RSA, and r is the R-peak timing-jitter variance. Both can be estimated from the recording or from a clean timing reference.

The deployed timing tracker is written as a PI phase-locked loop. We index the update by the R-peak that has just arrived: beat k provides the observed interval z_k , the update produces $\hat{T}_k$, and $\hat{T}_k$ is then used to predict beat $k+1$. During the interval preceding beat k , the phase advances using $\hat{T}_{k-1}$. At the detected sample i_k , define the pre-reset phase error as $e_{\phi,k} = \hat{\phi}(i_k^-) - 1$:

$$\hat{\phi}(i) = \hat{\phi}(i-1) + \frac{1}{f_s \hat{T}_{k-1}}, \quad (13a)$$

$$\hat{\phi} \leftarrow \hat{\phi} - K_p e_{\phi,k}, \quad (13b)$$

$$\hat{T}_k \leftarrow \hat{T}_{k-1} + K_{ss} (z_k - \hat{T}_{k-1}). \quad (13c)$$

At the detected fiducial i_k , the phase origin is reset to $\hat{\phi}(i_k) = 0$ before beat k is advanced using the newly updated $\hat{T}_k$. The steady-state gain K_{ss} is designed using the standard scalar Kalman filter associated with Eq. (11). The process variance q and measurement variance r may be calibrated offline from representative recordings or adapted online from previously accepted RR intervals. The resulting K_{ss} is then used in Eq. (13c); no separate Kalman recursion is required by the proposed cancellation method. In the present implementation, offline calibration is preferred because online q/r adaptation cannot reliably separate RR wander from R-peak timing jitter without a sufficiently long accepted-beat history.

K_p controls onset-jitter smoothing independently: values near one snap the phase to a detected fiducial, while smaller values smooth noisy onset corrections. The loop bandwidth should remain above the RSA band and below the per-beat detection-noise bandwidth.

Note 5. To prevent abnormal beats from corrupting the timing estimate or the repetitive template, we use the normalized RR innovation as a gating metric:

$$\nu_k = \frac{z_k - \hat{T}_k^-}{\sqrt{P_k^- + r}}. \quad (14)$$

Here P_k^- is the prior RR-estimation covariance maintained by the scalar Kalman implementation. When $|\nu_k|$ exceeds a threshold η , the beat is treated as unreliable: the R-peak predictor and template update may be frozen, or cancellation may fall back to buffered mode. This gate complements the existing RR-ratio and morphology checks.

Note 6. The streaming real-time implementation of the iterative cancellation lookup and subtraction require $O(1)$ work per sample and $O(n_t)$ memory. The scalar Kalman PI predictor proposed in this subsection is only a showcase of how to obtain a real-time phase-tracking mechanism that can be offloaded to the ECG/R-peak device supplying the timing reference.

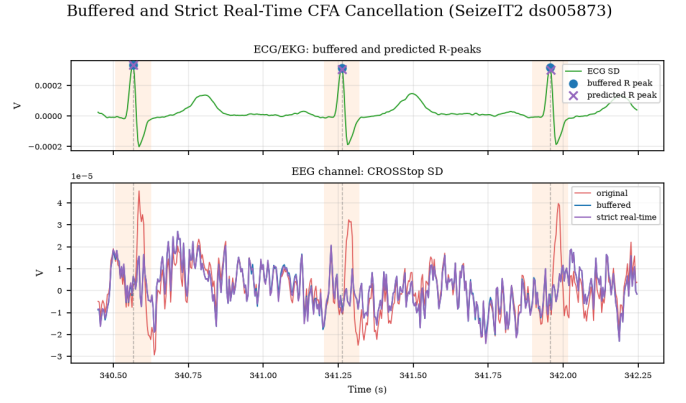


Fig. 1. Buffered and strict real-time repetitive CFA cancellation on a short SeizeIT2 segment (ds005873, sub-070, ses-01, run-20, behind-the-ear channel; 340.45–342.25 s). Top: chest ECG (green) with buffered detected R-peaks (blue circles) and predicted strict-real-time R-peaks (purple crosses). Bottom: original (red) and buffered (blue) and strict real-time (purple) ear-EEG. Both outputs use the repetitive template of Eq. (7); the buffered mode uses detected R-peaks, whereas the strict-real-time mode uses the predicted timing anchor.

IV. EXPERIMENTS

To evaluate the method we implemented a robust R-peak detector and PI phase tracker for both real and simulated data [13]. We demonstrate the RRO-template canceller only on records where our detector achieves reliable phase locking, and defer general-purpose phase tracking to future work. We likewise omit the strict-causal canceller below, as it hinges on timely R-peak prediction rather than on the template itself.

A. Real Ear-EEG Validation

We use the two-channel ear-EEG configuration of the SeizeIT2 dataset, a multicenter wearable epilepsy corpus of long-duration multimodal recordings (behind-the-ear EEG, ECG, EMG, ACC, GYR) with focal-seizure annotations across a large cohort [14]. Each EEG channel is high-pass filtered at 0.5 Hz, notch filtered at 50 Hz and resampled onto the ECG sample grid, and only supervisor-approved learnable beats enter the average. Because the 600 ms window can exceed the beat interval at high heart rates, overlapping template contributions are averaged before subtraction.

Fig. 1 shows buffered and strict-real-time repetitive disturbance rejection on the SeizeIT2 segment. The template is learned from $N_r = 851$ learnable beats over a 600 s window; the strict-real-time trace uses the predicted timing anchor and therefore exposes the timing-jitter penalty relative to the buffered trace.

Template SNR via a phase-scramble null. Since the true disturbance d_r is unobservable on real recordings, we validate the template against a null that shares its statistics but destroys R-phase locking, rather than against a direct error to d_r . Define the R-locked energy of a signal z under a peak set P ,

$$\mathcal{E}_P[z] = \left\| \frac{1}{|P|} \sum_{k \in P} z(t_k + p/f_s) \right\|_2^2, \quad (15)$$

evaluated over the main-lobe window around the R-peak (50 ms pre, 150 ms post, L_{lobe} samples), where the R-locked deflection is concentrated. Averaging on the true peaks gives the R-locked template, whereas a circular time-shift of the peak train, which preserves the RR-interval sequence but destroys absolute phase, gives leakage only. Averaging 200 such shifts yields the null floor $\mathcal{E}_{\text{scr}}$. Their ratio,

$$Q = \frac{\mathcal{E}_{\text{true}}[y]}{\mathcal{E}_{\text{scr}}[y]}, \quad \text{SNR}_{\text{templ}} = 10 \log_{10} Q, \quad (16)$$

is the CFA energy standing above the RRO-removed noise floor; since the same leakage enters both numerator and denominator, Q estimates $1 + \text{SNR}$, an offset negligible at the levels below. We further define a bias-corrected CFA RMS, a scale-free measure of the artifact the template removes:

$$A/\text{SD} = \frac{\sqrt{\max(\mathcal{E}_{\text{true}} - \mathcal{E}_{\text{scr}}, 0)/L_{\text{lobe}}}}{\text{SD}}. \quad (17)$$

The subtraction removes the leakage bias and $\text{SD} = \text{std}(y)$ is computed over the full preprocessed segment, so A/SD is the R-locked deflection in units of EEG standard deviation. For the real-data example in Fig. 1, the metric reads $\text{SNR}_{\text{templ}} = 28.80$ dB and $A/\text{SD} = 1.061$, indicating a strong and clearly resolvable R-locked component. From the whole dataset, we could observe $\text{SNR}_{\text{templ}}$ above 20 dB in 319 runs from 36 subjects, with $A/\text{SD} \in [0.110, 1.220]$ and mean $A/\text{SD} = 0.557$, as a showcase of the proposed method's effectiveness. Again, robust R-peak/beat-phase estimation on arrhythmic or low-quality ECG, and systematic characterization across the full dataset, are outside the scope of this work.

Coherence view. Fig. 2 contrasts the same R-locked ECG window against three EEG views: the raw ear-EEG beat, the stacked non-causal template $\hat{d}_r$, and the buffered one-beat-delay template. The left panel shows that the artifact is timing-repeatable: both templates lock to the same R-centered structure, and the buffered template is nearly indistinguishable from the non-causal one. The right panel shows that the templates also retain appreciable coherence with the ECG beat over the lower part of the band, while the raw-beat coherence is lower and less consistent, diluted by the neural background. Estimated from a single R-locked window, these curves are read as band structure rather than as coherence values.

The consequence is therefore not that the ECG waveform is uninformative, but that exploiting it is more demanding than exploiting timing. An ECG-referenced canceller must be driven by the waveform itself, sample-aligned to the EEG, whereas Eq. (7) consumes only the instants $\{t_k\}$ (Note 1). Both are available on a single instrument; across the device boundary targeted here, only the timing is. Sec. IV-B compares the two under controlled predictability, where ground truth is available.

B. Synthetic Simulation

We now sweep the ECG-to-artifact channel from linear (high coherence) to strongly nonlinear (near-zero coherence) at fixed artifact power. ICA-based CFA removal is effective given

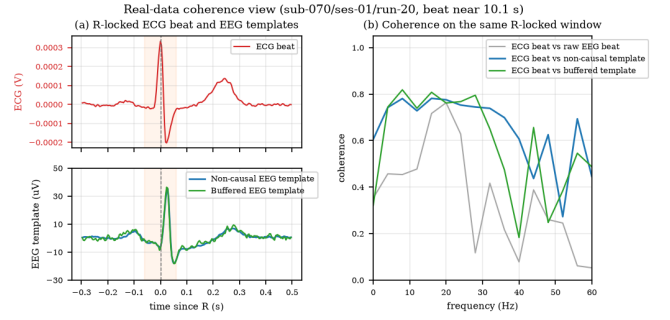


Fig. 2. Coherence view on the same real recording. Left: one R-locked ECG beat overlaid with the non-causal EEG template and the buffered one-beat-delay EEG template. Right: coherence between the windowed ECG beat and three R-locked EEG views: the raw EEG beat, the stacked non-causal template, and the buffered template. Both templates lock to the same R-centered structure and retain appreciable coherence with the ECG at lower frequencies, while the raw beat is diluted by the neural background. Estimated from a single R-locked window, these curves indicate band structure and are not used quantitatively.

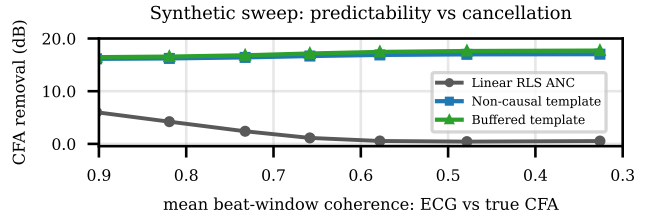


Fig. 3. Synthetic sweep of reference predictability versus cancellation performance. The sweep blends an ECG-coherent linear artifact with a phase-locked but waveform-decorrelated component at fixed artifact power; the x-axis is the mean 1–20 Hz coherence between the averaged ECG beat and the averaged *true* CFA beat, and the y-axis is SNR improvement over the contaminated input measured against the true neural signal. Linear RLS-ANC tracks coherence, whereas the non-causal and buffered template methods stay effective by relying on timing repeatability rather than waveform predictability.

sufficiently many channels but not in low-channel-count ear-ables, so we compare against a standard RLS reference-based canceller. Fig. 3 compares the linear ECG-referenced baseline (RLS-ANC) against the proposed rejection under controlled predictability. RLS-ANC acts as a coherence-gated canceller, suppressing the CFA only to the noise floor and degrading as ECG-to-CFA coherence falls. The repetitive methods remain stable, since they estimate the heartbeat-locked disturbance directly from the EEG using beat timing, and so remove nonlinear and coherence-diluted R-locked components as well as the linearly coherent part. Note that RLS-ANC is granted a sample-aligned ECG waveform throughout, i.e. its best case; the repetitive canceller matches it without requiring one.

V. CONCLUSION

We have presented a control-theoretic view of cardiac-field-artifact (CFA) removal in low-channel ear-EEG: CFA is modeled as a heartbeat-phase-locked repetitive disturbance and rejected by phase-indexed synchronous averaging and iterative learning, anchored only to R-peak timing and never to the

ECG waveform, so the reference may originate from a separate device that shares only beat instants. By improving SNR in the low-channel ear-EEG setting, it supports non-invasive wearable BCI applications such as brain-state monitoring and low-channel whole-brain modeling [15].

More broadly, this work aims to bridge mature mechatronic disturbance-rejection theory and biosignal denoising, opening a control-systems avenue for wearable neural interfaces. This perspective extends naturally beyond the cardiac artifact: an inertial (gyroscope/accelerometer) reference could drive feed-forward cancellation of motion artifact by the available sensors.

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